VoltBay presentation: who says what

Speaking guide for a team of six. Total talk time about 20 minutes, then questions. Replace "Member 1 to 6" with your names.

Member	Topic	Slides	Minutes on the clock	One-line job
Member 1	The idea and the storefront	1 to 4	0 to 4	Open the talk. Say what VoltBay is, then show the shop and the Gaming Arena.
Member 2	The foundation	5 and 6	4 to 7	Explain how it is built and how carts are remembered.
Member 3	The intelligence	7 and 8	7 to 10.5	Explain personalised picks and the VoltAI assistant.
Member 4	Buying and trust	9 and 10	10.5 to 13.5	Show checkout and pricing, then reviews and order tracking.
Member 5	The back office	11 and 12	13.5 to 16.5	Show the admin panel and how the owner runs the store.
Member 6	Quality and launch	13 to 15	16.5 to 20, then questions	Explain security, tests and going live. Close the talk and take questions.

RULES FOR EVERYONE

- Speak for about 1.5 minutes per slide. Do not read the slide; say it in your own words using the notes below.
- Each member ends with a hand-over line that names the next member.
- Say "demo store" when you talk about payments, prices and orders. Orders are pay on delivery and prices are indicative. We do not take real money.
- Photos come from Wikimedia Commons with their licences. A few generic products (for example the headset, monitor, chair and webcam) use representative photos and say so.

BEFORE THE PRESENTATION (MEMBER 6 CHECKS, EVERYONE HELPS)

- Open **https://voltbay-store.onrender.com** five minutes early. The free plan sleeps when idle and the first load can take up to a minute.
- Open the presentation file **VoltBay_Presentation_final.pptx** and keep one browser tab on the live site.
- Member 5 keeps the admin email and password ready, but never shows the password on screen.
- Have a backup: if the internet fails, use the screenshots in the slides. They are real pictures of the site.
- Test the microphone, and agree who clicks to the next slide (the speaker or one person for everyone).

Good to know: the site keeps carts and orders in memory on a free server. They reset whenever it restarts. That is fine for a demo, and Member 6 explains the fix under "what comes next".

1 Member 1: The idea and the storefront

Slides 1 to 4 | About 4 minutes | Opens the talk

SLIDE 1 AND 2: OPENING (30 SECONDS)

Say: "Good morning everyone. We are Team [name], and this is VoltBay, an online electronics store that we built and put on the internet. It remembers your cart, suggests products that fit you, and has an AI assistant that helps you shop. There are six of us, and each person will explain one part. Here is who covers what."

SLIDE 3: WHAT IS VOLTBAY? (1.5 MINUTES)

Say: "VoltBay sells phones, laptops, tablets, cameras, gaming gear, PC parts and home tech. It has 69 real products in 15 departments, and every product has a photo, a description and full specifications. Three things make it different from a basic shop: your cart is remembered when you come back, the shelf adapts to what you browse, and VoltAI can answer shopping questions using the real catalogue."

- 69 products, 15 departments, 2 AI modes (Gemini and an offline assistant).

SLIDE 4: STOREFRONT AND GAMING ARENA (1.5 MINUTES)

Say: "On the shop page you can search, sort by price, show only items in stock, save items to a wishlist and compare products side by side. The Gaming Arena groups consoles, controllers, VR, keyboards, mice, monitors, chairs, streaming gear and PC parts. It also has ready-made setups, and one click adds a whole setup to your cart."

SHOW ON THE LIVE SITE (30 SECONDS)

- Type "samsung" in search, then clear it. Click the heart on one product.
- Click **Gaming**, then **Add all to cart** on "Console starter". Say "Member 2 will explain how this cart is saved."

FACTS TO REMEMBER

- 14 gaming products and 10 gaming categories. Every photo has its author and licence in the product window.
- Products have a summary, highlights and a specification table.

IF SOMEONE ASKS

Are the products and prices real? The products are real, well-known models with their published specifications. Prices are indicative for a demo store.

Where do the photos come from? Wikimedia Commons. We store the author and licence and show them in the product window.

Hand over: "Now Member 2 will explain how all of this is built and how it remembers your cart."

2 Member 2: The foundation

Slides 5 and 6 | About 3 minutes

SLIDE 5: HOW VOLTBAY IS BUILT (1.5 MINUTES)

Say: "The browser talks to one small server written in Node.js. We did not install any extra packages. The server keeps everything in a built-in SQLite database. For the assistant it calls Google Gemini, and the product photos were downloaded once from Wikimedia with their licences. The browser asks the server for products through a simple REST API, and the shop and the admin panel use the same API."

- Node.js 22, no extra packages, built-in SQLite, one REST API.

SLIDE 6: CARTS THAT REMEMBER YOU (1.5 MINUTES)

Say: "When you first visit, your browser gets a private random token. The server stores only a scrambled version of it, called a hash, so even we cannot read your token. Your cart, wishlist and recently viewed products are saved in the database. If you come back within 90 days, the same cart is waiting. If you sign in, your guest cart is merged into your account, so nothing is lost."

SHOW ON THE LIVE SITE (30 SECONDS)

- Add a product to the cart, then press **F5** to reload. The cart is still there.

FACTS TO REMEMBER

- Token is random, and only its SHA-256 hash is stored. Sessions last 90 days and renew when used.
- Stock is checked on the server, never trusted from the page.
- Creating an account upgrades the guest in place, and signing in merges the guest cart into the account.

IF SOMEONE ASKS

Why SQLite and not MySQL or PostgreSQL? It needs no setup and is built into Node.js, so the project runs anywhere. The database code is in one file, so moving to PostgreSQL later is easy.

What if I clear my browser data? Then the token is gone and you start a new guest session. Signing in with an account keeps your cart on every device.

Hand over: "Now Member 3 will explain how VoltBay picks products for you and how the AI assistant works."

3 Member 3: The intelligence

Slides 7 and 8 | About 3.5 minutes

SLIDE 7: RECOMMENDATIONS THAT FIT YOU (1.5 MINUTES)

Say: "We only use what the shopper really does: products they view, what they search for, items they save or compare, what is in the cart, past orders, and a budget if they tell us one. Recent actions count more than old ones. Each product gets a score for category, brand, price closeness and accessories that go with what is in the cart. Every recommendation also shows a reason, such as 'Because you looked at Microsoft Xbox Series X'."

SHOW ON THE LIVE SITE (30 SECONDS)

- Open two or three gaming products, close them, go Home and scroll to **Recommended for you**. Read one reason aloud.

SLIDE 8: VOLTAI, OUR SHOPPING ASSISTANT (2 MINUTES)

Say: "VoltAI is our shopping assistant. It uses Google Gemini with ten tools: it can search, compare, recommend, view the cart, track an order, set your budget and remember preferences. Anything that changes your cart waits for you to press Confirm. We also protect the key with daily limits, and if the AI is slow or unavailable, an offline assistant answers instead, so the shopper never gets an error."

SHOW ON THE LIVE SITE (45 SECONDS)

- Click **Ask VoltAI** and type: "my budget is 30000, suggest a good phone".
- Point at the product cards in the answer and press **Add** to show the Confirm step.

FACTS TO REMEMBER

- Model: gemini-3.5-flash-lite. 10 tools. Limits: 200 AI requests a day for the whole store and 25 answers per shopper per day.
- General questions are reused for 3 hours, which saves quota. Personal questions are never reused.
- If the AI fails, the model is paused for a minute and the offline assistant answers straight away.

IF SOMEONE ASKS

Can the AI make up prices? No. Prices, stock and specifications come from the store database through the tools, and the instructions tell it not to invent them.

Does the AI buy things by itself? No. Cart changes always wait for the shopper to press Confirm.

What if Gemini is down or the limit is used up? The offline assistant takes over automatically.

Hand over: "Now Member 4 will show what happens when a customer actually buys something."

4 Member 4: Buying and trust

Slides 9 and 10 | About 3 minutes

SLIDE 9: CART, COUPONS AND CHECKOUT (1.5 MINUTES)

Say: "The cart drawer shows each item, a coupon box and add-ons such as warranty. The important part is that the server works out the final price, not the web page. Total equals subtotal minus coupon minus deals plus add-ons. This way the receipt, the saved order and the owner's revenue always match. Checkout also checks the stock and saves the order in one safe step, so two people cannot buy the last item."

SHOW ON THE LIVE SITE (45 SECONDS)

- Open the cart, type **VOLT10** and press **Apply** (10% off). Type a name and press **Place order**. Read the order number.

SLIDE 10: REVIEWS AND ORDER TRACKING (1.5 MINUTES)

Say: "Only a signed-in customer who really bought the product can review it, and only once per product, though they can edit it. The server checks the order history, so fake reviews are blocked. The stars on product cards are real averages of the stored reviews. After an order, the customer can see its status, which moves from Pending to Packed, Shipped and Completed, and can also ask VoltAI where their order is."

FACTS TO REMEMBER

- VOLT10 is 10% off. Four products have deals (10, 20, 15 and 25 percent). Delivery is free above 999 rupees.
- Payment is pay on delivery. This is a demo store with no online payment.
- Order number on the receipt is the same number the admin sees.

IF SOMEONE ASKS

Can I pay online? Not yet. Online payments and email confirmations are our first next step.

Why does the server calculate the price? The web page can be changed by a user. The server cannot, so it is the only safe place to decide the price.

Hand over: "Now Member 5 will show what the shop owner sees behind the scenes."

5 Member 5: The back office

Slides 11 and 12 | About 3 minutes

SLIDE 11: ADMIN PANEL, SIGN-IN ONLY (1.5 MINUTES)

Say: "Shoppers never see an Admin menu. This is enforced by the server, not just hidden: every admin request without an admin sign-in is refused, and the admin script itself is only sent after the admin signs in. The overview shows real numbers: revenue from stored orders, stock value, items that need restocking, and sales by category."

SHOW ON THE LIVE SITE (45 SECONDS)

- Click **Sign in** and enter the admin email and password. Do not show the password on screen; type it quickly or have it pasted.
- Open **Admin, Manage store** and show the Overview tab.

SLIDE 12: RUNNING THE STORE (1.5 MINUTES)

Say: "The owner can change an order's status, edit price and stock, add products, bill walk-in customers at the counter, and remove a review that breaks the rules. Repair bookings made by customers arrive here too, and the owner updates their status and estimate. The AI control tab shows how many AI requests were used today, has a kill switch for VoltAI, and lets the owner switch each tool on or off."

FACTS TO REMEMBER

- Tabs: Overview, Inventory, Counter billing, Orders, Reviews, Repairs, Activity feed, AI control, VoltAI Command, VoltShield.
- The activity feed is a real audit log of admin actions and AI tool calls.
- VoltAI Command and VoltShield are simple rule-based demos, not real fraud or AI monitoring. Say this if asked.

IF SOMEONE ASKS

Can a customer open the admin page by guessing the address? No. The server answers 401 or 403 to everyone except the signed-in admin.

Do admin edits stay? Yes, they are saved in the database. On the free server they reset if it restarts.

Hand over: "Finally, Member 6 will explain how we kept it safe, tested it, and put it online."

6 Member 6: Quality and launch

Slides 13 to 15 | About 3.5 minutes | Closes the talk

SLIDE 13: SECURITY AND TESTING (1.5 MINUTES)

Say: "Security first. Passwords are hashed with scrypt, so they are never stored as text. Login attempts are limited, session tokens are stored hashed, private files such as our settings file and the database cannot be downloaded, and the server sends strict browser security headers. For testing we have 21 automated server tests and 9 real-browser tests. The browser tests open a real Chrome, click through the shop, check out, sign in as admin and post a review. All of them pass."

SLIDE 14: GOING LIVE AND WHAT COMES NEXT (1.5 MINUTES)

Say: "The code is on GitHub. A Dockerfile packages the app, and Render builds it and runs it on the internet. The AI key and the admin password are stored as private settings on Render, never in the code. On the free plan the site sleeps when nobody visits, and its data resets when it restarts. Next we would add online payments, email confirmations, a permanent database, and more real products, especially monitors."

SLIDE 15: CLOSE AND QUESTIONS (30 SECONDS)

Say: "Thank you. You can open the live site at voltbay-store.onrender.com and find our code on GitHub at nskpr2008 slash Python-Project. We are happy to take your questions."

FACTS TO REMEMBER

- Tests: 21 API tests and 9 browser tests. We also broke the admin lock on purpose once to check that the tests catch it.
- Hosting: Render free plan, Singapore region. First load after sleeping can take up to a minute.
- Secrets (the Gemini key and admin password) live only in Render's settings and a private file on the laptop.

IF SOMEONE ASKS

Is it safe to put on the internet? The basics are covered (hashing, limits, locked admin, private files). Before real customers we would add HTTPS-only cookies and real payments from a payment provider.

Why does data reset? The free server has no permanent disk. A paid disk or an external database fixes it.

Questions and answers for everyone

Anyone can answer these. If you do not know, say "Member N can answer that" and name the right person.

Question	Short answer	Who
Which technology did you use?	Node.js 22 with the built-in SQLite database, plain HTML and JavaScript in the browser, and Google Gemini for the assistant. No extra packages.	Member 2
How does the AI know your products?	It uses tools that search the store database, so answers come from real prices, stock and specifications.	Member 3
What stops the AI from running up a big bill?	Daily limits: 200 AI requests a day for the store and 25 per shopper. After that, the offline assistant answers.	Member 3
How do recommendations work?	They score products by what the shopper viewed, searched, saved and bought, with recent actions counting more, and show a reason on each card.	Member 3
Can someone cheat the price or stock?	No. The server calculates prices and checks stock when the order is placed.	Member 4
Can anyone write a fake review?	No. Only a signed-in customer who bought the product can review it, once.	Member 4
How is the admin panel protected?	Every admin request needs an admin sign-in, and the admin script is only sent to the admin. Visitors get 401 or 403.	Member 5
How do you know it works?	21 server tests and 9 real-browser tests pass, and we check the live site after every update.	Member 6
Why is the first load slow sometimes?	The free hosting plan sleeps when idle and takes up to a minute to wake up.	Member 6
Is this a real shop?	It is a working demo store. Products and specs are real, prices are indicative, and payment is pay on delivery with no real money.	Member 1
What would you add next?	Online payments, email confirmations, a permanent database, more real products (especially monitors) and real fraud checks.	Member 6
What was the hardest part?	Keeping the price, stock and cart the same everywhere. We solved it by letting the server own them, and we wrote tests so a change cannot break it quietly.	Member 4